Saving, Investment, and the Financial System (chap 26)

Money, Growth and Inflation (chap 30)

Financial Institutions

- Financial system
  - Group of institutions in the economy
    - That help match one person's saving with another person's investment
  - Moves the economy's scarce resources from savers to borrowers
- Financial institutions
  - Financial markets
  - Financial intermediaries

Financial Markets, Part 1

- Financial markets
  - Savers can directly provide funds to borrowers
  - The bond market
  - The stock market

Financial Markets, Part 2

- The bond market
  - Bond: certificate of indebtedness
    - Date of maturity, when the loan will be repaid
    - Rate of interest, paid periodically until the date of maturity
    - Principal, amount borrowed
  - Borrowing from the public
    - Used by large corporations, the federal government, or state and local governments

Financial Markets, Part 3

- The bond market
  - Term: length of time until maturity
    - A few months, 30 years, perpetuity
    - Long-term bonds are riskier than short-term bonds
      - Long-term bonds usually pay higher interest rates

Financial Markets, Part 4

- The bond market
  - Credit risk: probability of default
    - Probability that the borrower will fail to pay some of the interest or principal
    - Higher interest rates for higher probability of default
    - U.S. government bonds tend to pay low interest rates
    - Junk bonds, very high interest rates
      - Issued by financially shaky corporations

Financial Markets, Part 5

- The bond market
  - Tax treatment
    - Interest on most bonds is taxable income
    - Municipal bonds
      - Issued by state and local governments
      - Owners are not required to pay federal income tax on the interest income
      - Lower interest rate

Financial Markets, Part 6

- The stock market
  - Stock: claim to partial ownership in a firm
    - A claim to the profits that a firm makes
  - Organized stock exchanges
    - Stock prices: demand and supply
  - Equity finance
    - Sale of stock to raise money
  - Stock index
    - Average of a group of stock prices

Financial Intermediaries, Part 1

- Financial intermediaries
  - Savers can indirectly provide funds to borrowers
  - Banks
  - Mutual funds

Financial Intermediaries, Part 2

- Banks
  - Take in deposits from savers
    - Banks pay interest
  - Make loans to borrowers
    - Banks charge interest
  - Facilitate purchasing of goods and services
    - Checks: medium of exchange

Financial Intermediaries, Part 3

- Mutual funds
  - Institution that sells shares to the public
  - Uses the proceeds to buy a portfolio of stocks and bonds
  - Advantages
    - Diversification; professional money managers

National Income Accounts

- Rules of national income accounting
  - Important identities
- Identity
  - An equation that must be true because of the way the variables in the equation are defined
  - Clarify how different variables are related to one another

Accounting Identities, Part 1

- Gross domestic product (GDP)
  - Total income
  - Total expenditure
- $Y = C + I + G + NX$
  - Y = gross domestic product, GDP
  - C = consumption
  - I = investment
  - G = government purchases
  - NX = net exports

Accounting Identities, Part 2

- Closed economy
  - Doesn't interact with other economies
  - NX = 0
- Open economy
  - Interact with other economies
  - NX ≠ 0

Accounting Identities, Part 3

- Assume closed economy: NX = 0
  - Y = C + I + G
- National saving (saving), S
  - Total income in the economy that remains after paying for consumption and government purchases
  - Y – C – G = I
  - S = Y – C – G
  - S = I

Accounting Identities, Part 4

- T = taxes minus transfer payments
  - S = Y – C – G
  - S = (Y – T – C) + (T – G)
- Private saving, Y – T – C
  - – Income that households have left after paying for taxes and consumption
- Public saving, T – G
  - – Tax revenue that the government has left after paying for its spending

Accounting Identities, Part 5

- Budget surplus: T – G > 0
  - Excess of tax revenue over government spending
- Budget deficit: T – G < 0
  - Shortfall of tax revenue from government spending

Saving and Investing

- Accounting identity: S = I
- Saving = Investment
  - For the economy as a whole
  - One person's savings can finance another person's investment

The Market for Loanable Funds, Part 1

- Market for loanable funds
  - Market
    - Those who want to save supply funds
    - Those who want to borrow to invest demand funds
  - One interest rate
    - Return to saving
    - Cost of borrowing
  - Assumption
    - Single financial market

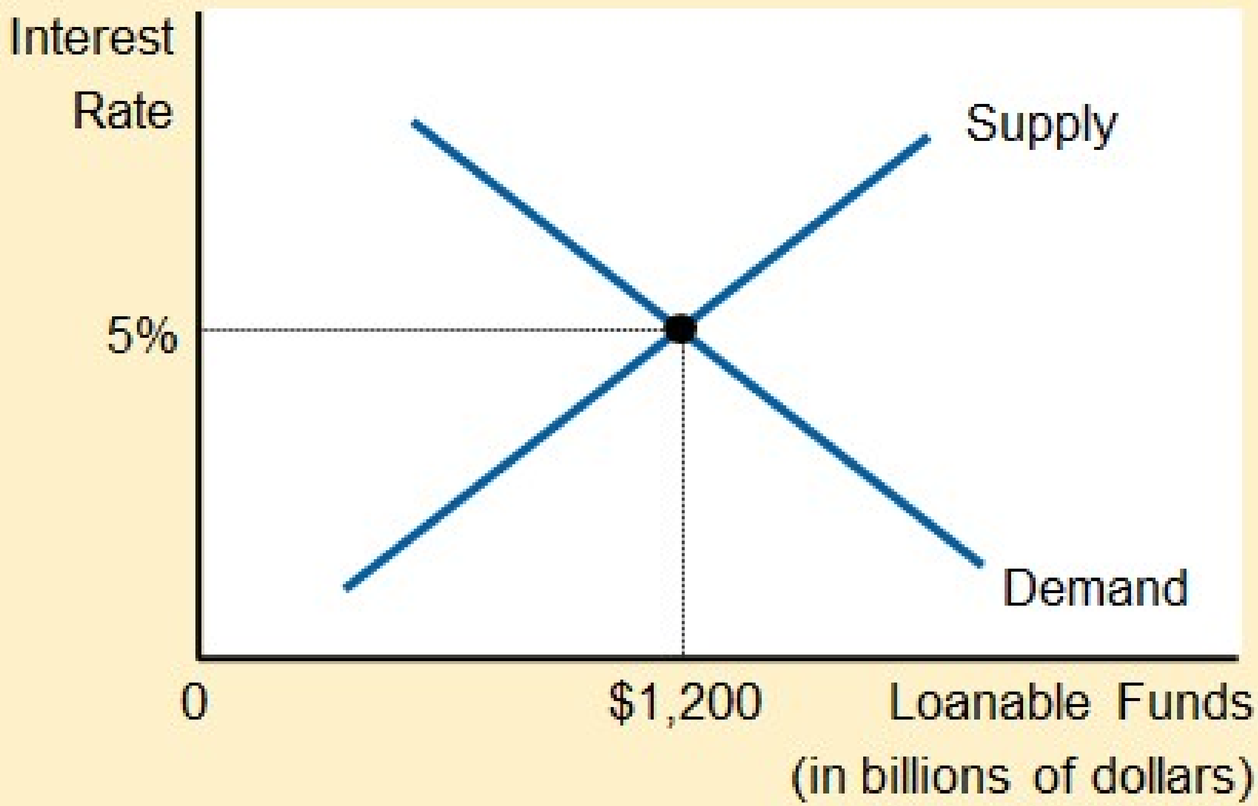
The Market for Loanable Funds, Part 2

- Supply and demand of loanable funds
  - Source of the supply of loanable funds
    - Saving
  - Source of the demand for loanable funds
    - Investment
  - Price of a loan = real interest rate
    - Borrowers pay for a loan
    - Lenders receive on their saving

The Market for Loanable Funds, Part 3

- Supply and demand of loanable funds
  - As interest rate rises
    - Quantity demanded declines
    - Quantity supplied increases
  - Demand curve
    - Slopes downward
  - Supply curve
    - Slopes upward

Figure 1
The Market for Loanable Funds



The interest rate in the economy adjusts to balance the supply and demand for loanable funds. The supply of loanable funds comes from national saving, including both private saving and public saving. The demand for loanable funds comes from firms and households that want to borrow for purposes of investment. Here the equilibrium interest rate is 5 percent, and $1,200 billion of loanable funds are supplied and demanded.

The Market for Loanable Funds, Part 4

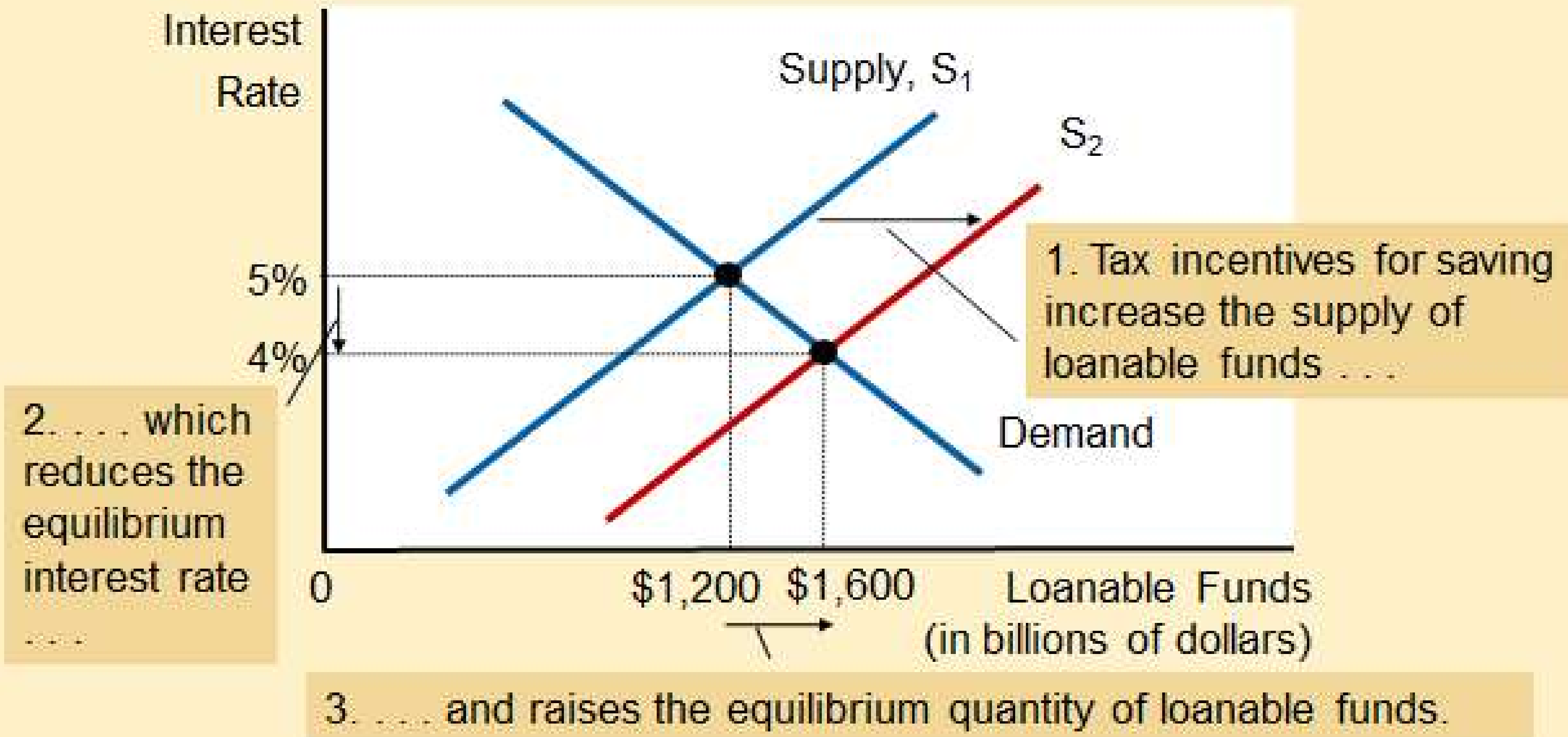
- Government policies
  - Can affect the economy's saving and investment
    - Saving incentives
    - Investment incentives
    - Government budget deficits and surpluses

Policy 1: Saving Incentives

- Shelter some saving from taxation(tax law that encourages savings: HH use this funding to buy more funds or increase their deposits in banks).
  - Affect supply of loanable funds
  - Increase in supply
    - Supply curve shifts right
  - New equilibrium
    - Lower interest rate
    - Higher quantity of loanable funds
      - Greater investment

Figure 2

Saving Incentives Increase the Supply of Loanable Funds

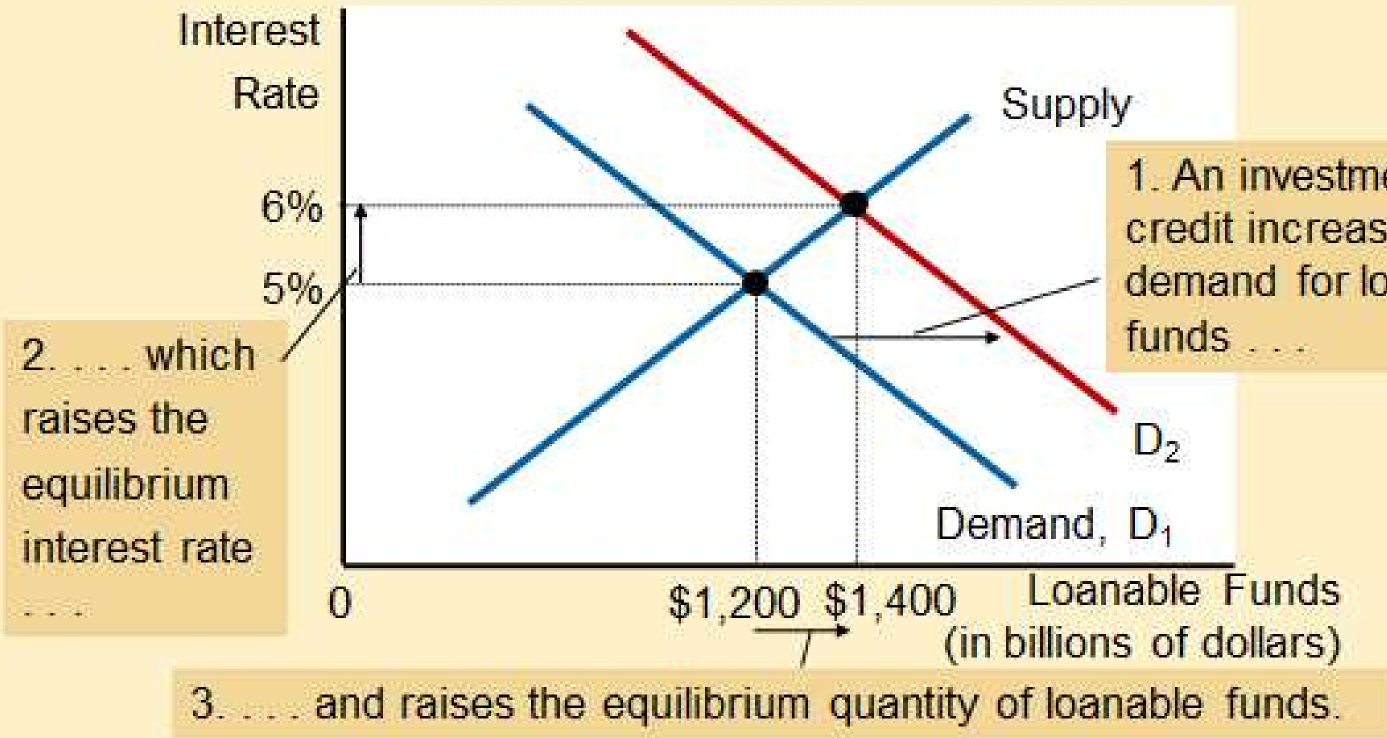


A change in the tax laws to encourage Americans to save more would shift the supply of loanable funds to the right from S_1 to S_2. As a result, the equilibrium interest rate would fall, and the lower interest rate would stimulate investment. Here the equilibrium interest rate falls from 5 percent to 4 percent, and the equilibrium quantity of loanable funds saved and invested rises from $1,200 billion to $1,600 billion.

Policy 2: Investment Incentives

- Investment tax credit(gives a tax advantage to any firms building a new factory or buying a new equipment)
  - Affect demand for loanable funds (reward firms that borrow and invest in new capital)
  - Increase in demand
    - Demand curve shifts right
  - New equilibrium
    - Higher interest rate
    - Higher quantity of loanable funds
      - Greater saving

Figure 3
Investment Incentives Increase the Demand for Loanable Funds



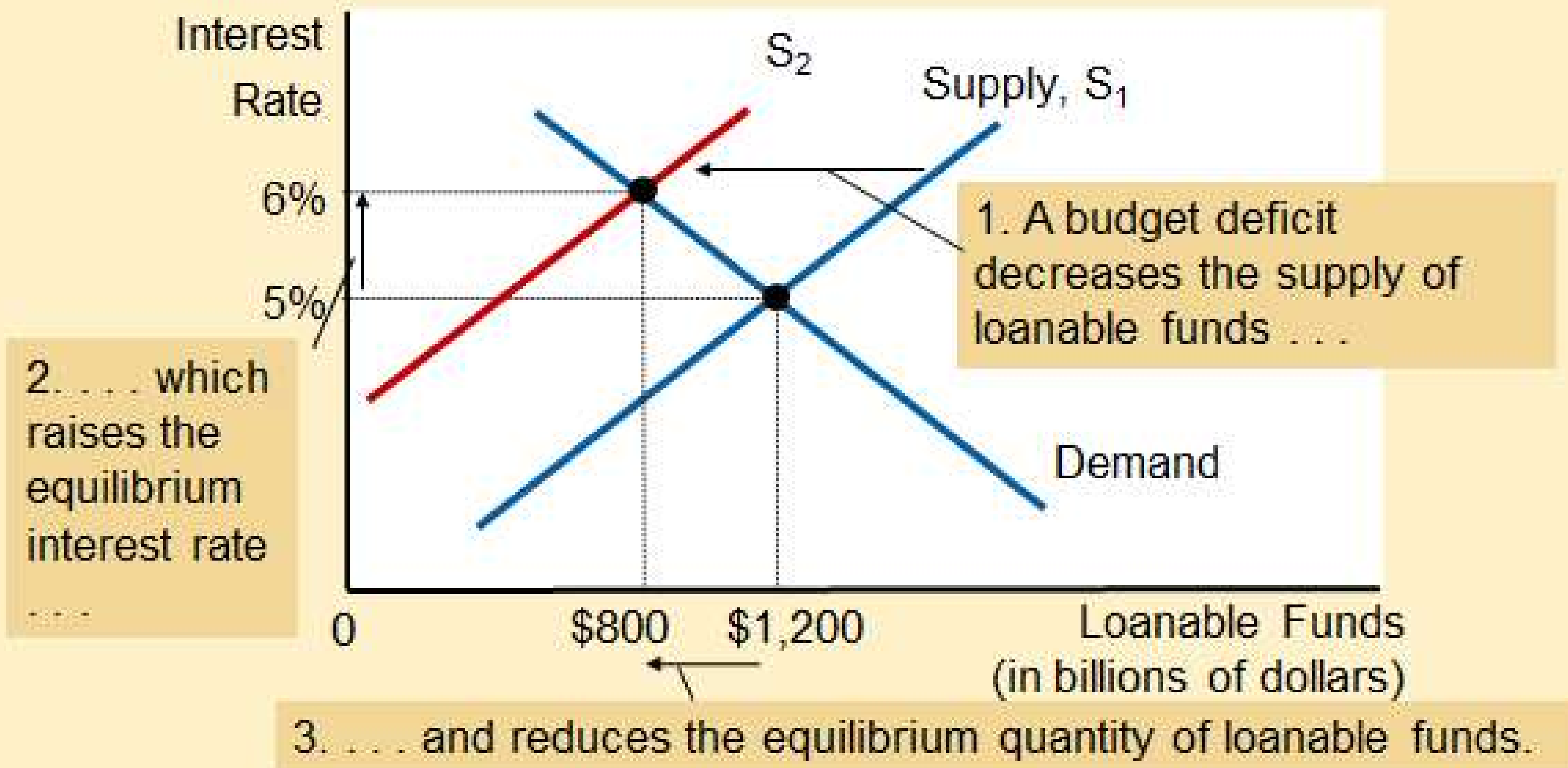
If the passage of an investment tax credit encouraged firms to invest more, the demand for loanable funds would increase. As a result, the equilibrium interest rate would rise, and the higher interest rate would stimulate saving. Here, when the demand curve shifts from D_1 to D_2, the equilibrium interest rate rises from 5 percent to 6 percent, and the equilibrium quantity of loanable funds saved and invested rises from \$1,200 billion to \$1,400 billion.

Policy 3: Budget Deficit/Surplus, Part 1

- Government - starts with balanced budget
  - Then starts running a budget deficit
    - Change in supply of loanable funds
    - Decrease in supply
      - Supply curve shifts left
    - New equilibrium
      - Higher interest rate
      - Smaller quantity of loanable funds

Figure 4

The Effect of a Government Budget Deficit



When the government spends more than it receives in tax revenue, the resulting budget deficit lowers national saving. The supply of loanable funds decreases, and the equilibrium interest rate rises. Thus, when the government borrows to finance its budget deficit, it crowds out households and firms (decrease inv because of govt borrowings) that otherwise would borrow to finance investment. Here, when the supply shifts from S_1 to S_2, the equilibrium interest rate rises from 5 to 6 percent, and the equilibrium quantity of loanable funds saved and invested falls from $1,200 billion to $800 billion.

Policy 3: Budget Deficit/Surplus

- Crowding out
  - Decrease in investment
  - Results from government borrowing
- Government - budget deficit
  - Interest rate rises
  - Investment falls

Policy 3: Budget Deficit/Surplus, Part 3

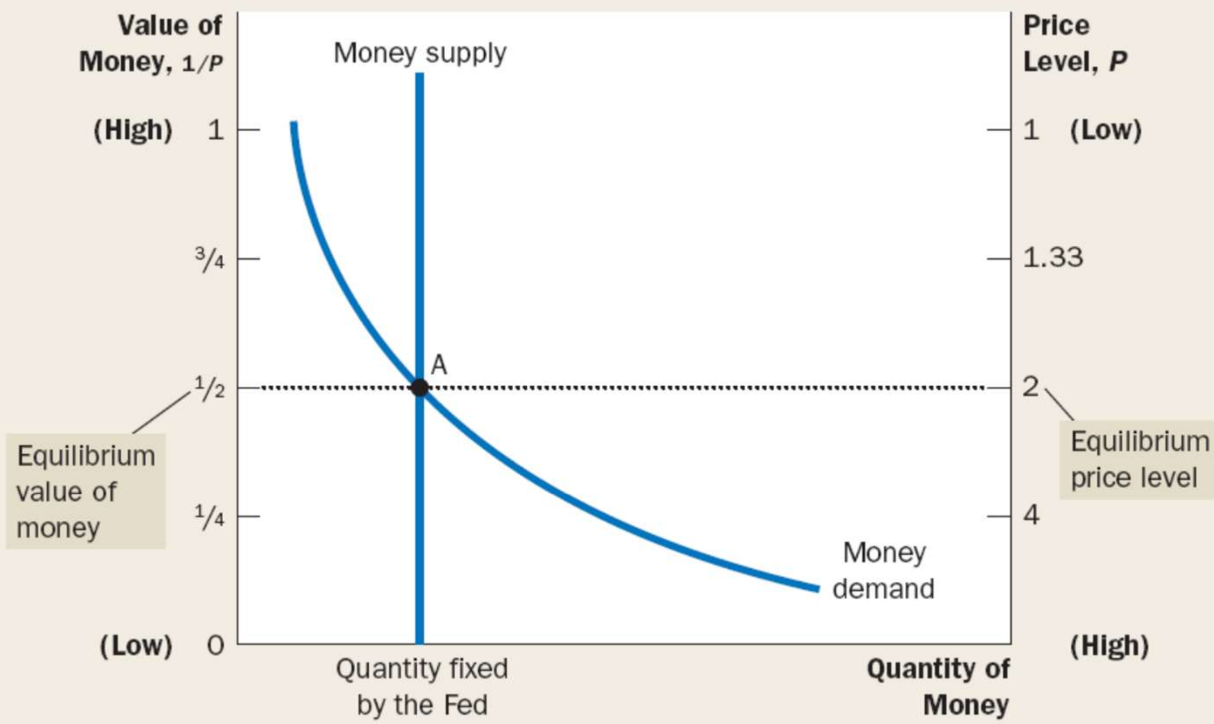
- Government – budget surplus
  - Increase supply of loanable funds
  - Reduce interest rate
  - Stimulates investment

Money Growth and Inflation (chap 30)

The horizontal axis shows the quantity of money. The left vertical axis shows the value of money, and the right vertical axis shows the price level. The supply curve for money is vertical because the quantity of money supplied is fixed by the Fed. The demand curve for money is downward sloping because people want to hold a larger quantity of money when each dollar buys less. At the equilibrium, point A, the value of money (on the left axis) and the price level (on the right axis) have adjusted to bring the quantity of money supplied and the quantity of money demanded into balance.

FIGURE 1

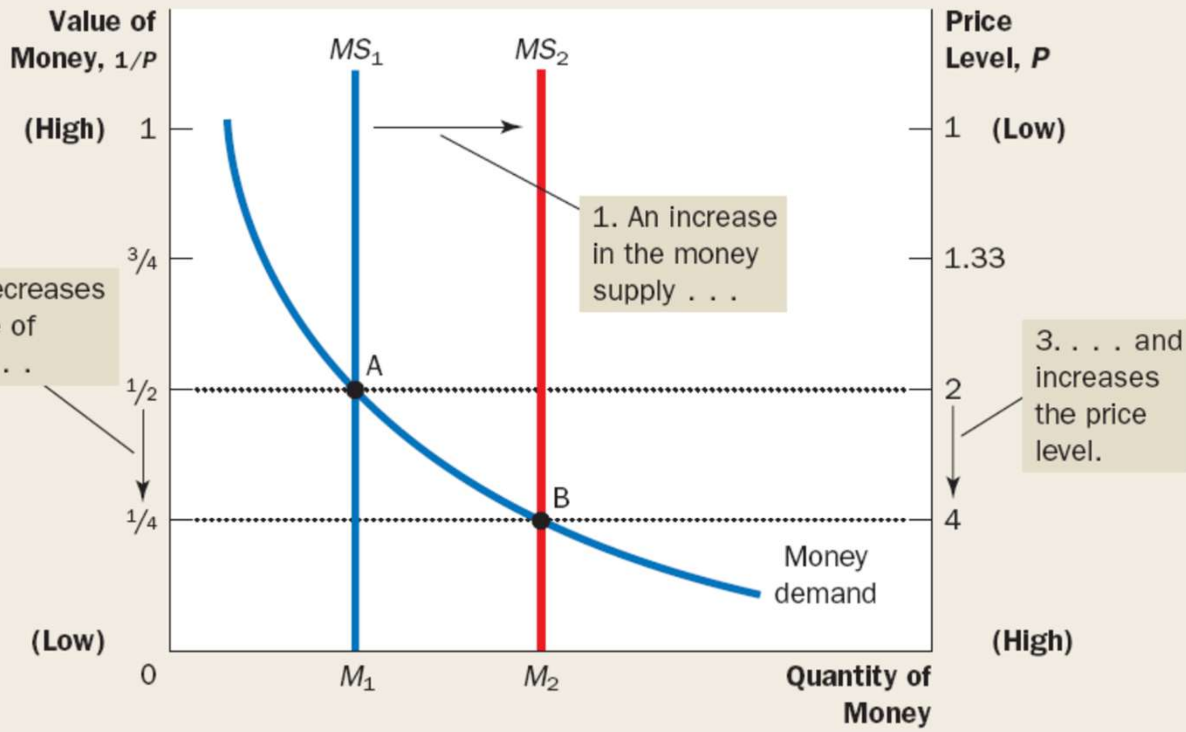
How the Supply and Demand for Money Determine the Equilibrium Price Level



2 FIGURE

An Increase in the Money Supply

When the Fed increases the supply of money, the money supply curve shifts from MS_1 to MS_2. The value of money (on the left axis) and the price level (on the right axis) adjust to bring supply and demand back into balance. The equilibrium moves from point A to point B. Thus, when an increase in the money supply makes dollars more plentiful, the price level increases, making each dollar less valuable.



This figure shows the nominal value of output as measured by nominal GDP, the quantity of money as measured by M2, and the velocity of money as measured by their ratio. For comparability, all three series have been scaled to equal 100 in 1960. Notice that nominal GDP and the quantity of money have grown dramatically over this period, while velocity has been relatively stable.

FIGURE 3

Nominal GDP, the Quantity of Money, and the Velocity of Money

Source: U.S. Department of Commerce; Federal Reserve Board.